

Project report on
Daily Dashboard



Undertaken at:
Noida

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Sopra Steria Ltd.** for providing me the opportunity to work as an intern during the period **September 2025 to November 2025**. This internship has been a valuable learning experience where I gained practical exposure to real-time data analytics and industry-level development workflows.

I extend my heartfelt thanks to my mentors and the entire project team for their continuous guidance, support, and constructive feedback throughout the development of the **Daily Dashboard – Real-Time Data Analytics Application**. Their insights greatly helped me enhance my technical skills and understand professional best practices.

I am also thankful to the **Department of Electronics and Communication Engineering, MIET, Meerut**, for providing me with the necessary resources and encouragement to undertake this project.

Finally, I would like to express my gratitude to my family and friends for their constant motivation and support during this journey.

CERTIFICATE

This is to certify that **Govind Sharma**, Roll No. **2200680310042**, a student of **B.Tech in Electronics and Communication Engineering**, MIET, Meerut, has successfully completed his internship titled:

“Daily Dashboard – Real-Time Data Analytics Application”

at **Sopra Steria Ltd.** during the period **September 2025 to November 2025**.

During this internship, he developed a real-time data analytics dashboard and worked on backend development, database management, automation scripts, and UI integration. His performance, discipline, and sincerity throughout the internship period were commendable.

This certificate is issued as a recognition of his successful completion of the internship and project work.

(Signature)

Industry Mentor

Sopra Steria Ltd.

About Sopra Steria Group

Sopra Steria is a leading European information technology consultancy that provides consulting, digital services, and software development to clients worldwide. The company employs over 50,000 professionals and serves diverse industries such as the public sector, finance, energy, and telecommunications. Sopra Steria focuses on transforming businesses through innovative technology solutions by emphasizing collaboration, agility, and sustainable development. During my internship, I had the opportunity to work alongside industry experts in a supportive and innovative environment, which greatly contributed to the successful development of my Daily Dashboard project.

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Abstract

The **Daily Dashboard – Real-Time Data Analytics Application** is a web-based solution developed to simplify data monitoring, automate reporting, and enhance decision-making within an organizational environment. The system integrates backend data processing, database management, and interactive visualizations to provide users with an intuitive and reliable platform for tracking daily operational metrics.

The application is built using **Flask (Python)** for backend logic, **SQLite and PostgreSQL** for lightweight and efficient data storage, and **HTML/CSS/JavaScript** for the user interface. It allows secure user login, real-time data updates, and automated PDF report generation. The dashboard displays key performance indicators (KPIs), charts, and tabular summaries, enabling quick insights into daily activities.

This project was developed during the internship at **Sopra Steria Ltd.** and reflects practical exposure to real-world data analytics workflows, application design, and software development best practices. The implementation of this system demonstrates the ability to build scalable, modular, and user-centric applications suited for professional environments.

Hardware & Software used

Hardware Used :

- Standard Laptop/Workstation
- Internet Connectivity

Software Used :

1. Python

Used for:

Data extraction

Processing & cleaning

Automation scripts

Backend logic

2. VS Code / PyCharm

Used as the main IDE to write, debug, and organize Python code.

3. FastAPI / Flask

Used for:

Creating backend APIs

Sending processed data to the dashboard

4. PostgreSQL / MySQL / SQL Database

Used to store:

Raw data

Processed data

Daily metrics

Logs

Introduction

Development & Programming

Python

-Used for data processing, automations, backend logic, and handling APIs.

Bootstrap

-Implemented dynamic elements and interactive features in the dashboard.

Frameworks & Libraries

Dash / Flask – web framework for building the dashboard

Database & Storage

MySQL / SQL Server

-For storing and retrieving processed and raw data.

Internal API services from Sopra Steria

-Used to fetch real-time data for analytics.

Development Tools

VS Code – primary code editor

Postman – testing API endpoints

Purpose of Project :

The primary purpose of this project was to **build a real-time, automated, and easy-to-use dashboard system** that allows teams to **track daily operational data**, monitor performance, and make faster, data-driven decisions.

Here's the purpose broken down clearly:

1. Automate Daily Data Processing

- Before this project, data extraction or reporting might have been manual.
Your dashboard solves that by:
- Automatically running Python scripts

2. Centralize All Insights in One Place

- Instead of working with scattered Excel sheets, files, or raw reports, the dashboard gives a **single centralized view** of:
- Daily metrics
- Weekly trends
- Monthly insights

3. Improve Decision-Making

- Because of real-time data and visualizations, teams can now:
- Identify bottlenecks
- Monitor performance
- Take informed actions faster

4. Create a Scalable, Reusable System

- Your architecture (Python + APIs + database + dashboard UI) is:
- Modular
- Maintainable
- Easy to expand for future requirements
- This makes it useful not just for your internship, but also future teams.

Scope of this Project :

The scope of this project involves designing and developing a fully automated Daily Dashboard system capable of handling end-to-end data processing. This includes extracting raw data from defined sources, cleaning and transforming it using Python scripts, and updating it in real time through scheduled automation. The project also covers building APIs to enable secure and efficient data retrieval, as well as integrating these APIs with a dashboard interface for visual representation of daily performance metrics.

The scope further extends to implementing key dashboard features such as trend analysis, date filtering, comparative insights, and performance summaries. It ensures that the system reduces manual effort in daily reporting, enhances data accuracy, and provides a centralized platform for monitoring operational activities. Additionally, the project is structured to be scalable, allowing new datasets, KPIs, or modules to be added easily in the future, making it adaptable to evolving business needs.

Description :

As a Python Developer Intern, I was responsible for designing and building the entire backend workflow for the Daily Dashboard project. The development process followed multiple structured steps:

1. Understanding Requirements

The first step was to understand what the team needed:

- A dashboard that updates daily without manual effort
- A backend system that automates data extraction
- APIs that send processed data to the dashboard
- A clean structure that can be scaled in the future
- I analyzed how the existing data sources worked and what KPIs needed to be shown.

2. Designing the Backend Architecture

Once requirements were clear, I planned the entire system architecture:

- **Python scripts** for extraction, processing, validation
- **Schedulers** for automation
- **Database tables** for raw, processed, and daily metrics
- **API layer** to provide data to the dashboard
- **Logging module** to track errors and performance
- This design ensured a reliable and maintainable backend.

3. Building Python Scripts for Data Extraction

I wrote Python modules to:

- Connect to APIs / files / internal sources
- Fetch raw data automatically
- Handle exceptions and log errors
- Store the data inside the `raw_data` table
- This was the foundation of the pipeline.

4. Cleaning and Transforming the Data

Next, I created processing scripts:

- Cleaned missing or incorrect values
- Applied transformations and rules
- Validated each record
- Stored the cleaned results in the `processed_data` table
- This step ensured accuracy before displaying data.

5. Automating the Entire Process

To make the dashboard *truly daily*, I automated everything using:

- **Scheduled Python jobs** (cron/scheduler)
- Automated refresh logic
- Error and retry handling
- This allowed the system to run without manual involvement.

6. Creating Backend APIs

I developed APIs using Python (FastAPI/Flask) to:

- Fetch daily metrics
- Provide processed data to the frontend
- Return responses in JSON format
- Handle errors and validations
- Deliver fast and optimized results
- These APIs act as the bridge between Python backend and the dashboard UI.

7. Designing Database Structure

I created database tables for:

- Raw data
- Processed data
- Daily metrics
- API logs
- User activity
- Each table had a clear purpose in the pipeline.

8. Error Handling, Logging & Debugging

I implemented:

- Log files for extraction, processing, API calls
- Error capturing
- Debugging routines
- Fail-safe mechanisms
- This made the system stable and production-ready.

9. Connecting APIs to the Dashboard

Once the backend was ready:

- API responses were consumed by the dashboard
- The dashboard displayed daily KPIs, charts, and trends
- Data updated automatically every day

This completed the end-to-end system.

10. Testing & Optimization

I tested:

- API speed
- Script accuracy
- Data reliability
- Scheduler stability
- Then I optimized the code for:
- Faster processing
- Reduced memory usage

Cleaner modular structure

11. Final Deployment & Documentation

I deployed the backend modules and created documentation covering:

- System architecture
- API endpoints
- Database schemas
- Automation flow
- This ensured the project could be maintained easily in the future.

Application of the Project :

- Real-time Performance Monitoring**

Allows teams to track daily operational performance without manual reporting.

- Automated Data Processing**

Eliminates repetitive data cleaning and updating tasks through Python automation.

- Centralized Information Access**

Provides a single dashboard for viewing all important KPIs and metrics.

- Decision-Making Support**

Helps managers quickly analyze trends and take timely decisions based on updated data.

- Improved Workflow Efficiency**

Reduces human errors and speeds up daily reporting cycles.

- **Scalable Backend System**

Supports adding new datasets, KPIs, or modules without major rework.

- Daily Insights for Teams**

Useful for operations, leadership, QA team **API-Based Data Sharing**

Allows other tools or teams to consume metrics through backend APIs created in Python.

Conclusion

The development of the *Daily Dashboard Project* allowed me to apply and strengthen my skills as a **Python Developer Intern**. Through this project, I worked extensively with Python for data handling, automation, API integration, and backend logic. I also gained practical experience in building a functional web-based dashboard using frameworks like Flask/Dash, along with JavaScript for frontend interactivity.

This project helped me understand how real-time systems function inside an organization, how APIs deliver continuous data, and how Python can be used to automate processes, structure workflows, and build scalable components. Overall, the project improved my problem-solving ability, coding standards, debugging skills, and understanding of real-world software development practices. It also gave me hands-on exposure to development tools, version control, and working in an internship environment.

References

- Python Official Documentation
- FastAPI / Flask Documentation
- PostgreSQL / SQL Reference Guides
- Visual Studio Code Documentation
- Git & GitHub Documentation
- Internal project documents and guidance from the internship
- Standard Python libraries and modules reference (Pandas, Requests, JSON, etc.)
- General software engineering best practices followed during development

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